Doppler-Aided Link Adaptation for Underwater Acoustic Links

Main idea, gaps, methodology and theoretical objectives — for group discussion

8 October 2026

Target: IEEE JSAC special issue on Next-Generation Underwater Acoustic Communication Systems (deadline 31 March 2027)

CLAIM stated in the idea document, not yet established SUGGESTED design option added after review OPEN needs material, a proof or an experiment SOURCE found by targeted search, not yet checked

Main idea

The proposal in one slide

Every packet, data or ACK, carries a free measurement of the Doppler scale $\epsilon = v_{\mathrm{rel}}/c$, the same at both ends of a common path. Both ends fuse them into one kinematic forecast of the drift $d = \dot{\epsilon}$ and its uncertainty, which drives three blocks:

- **frame**: pilots placed where the forecast is uncertain; packet length, modulation
- **receiver**: compensation pre-warped along the forecast, tracking only its error
- **learner**: a contextual bandit that calibrates PER online under delayed ACKs

Candidate claim SUGGESTED: One kinematic forecast, shared by both ends, drives frame design and receiver compensation, calibrated online; PER control is a safety layer.

Scope

Delay-tolerant traffic
Mobile or surface-coupled nodes
Seas with a swell component
Paths resolvable in delay or Doppler

Still, fixed links: fall back to the baseline; "no harm" once overheads are counted is still to be shown.

Gaps with the literature

Related work (1): adapting the link despite stale feedback

Line of work	Representative papers	Difference (abstract level)
Physics-informed, uncertainty-aware AMC (closest)	Anjangi et al., UComms 2018 (Doppler spread as AMC feature); Wu, Chitre, Anjangi, JOE 2025 (GP interval BER + feedback scheduling)	MCS only; no kinematic forecast, packet layout or timing
Channel prediction for AMC	Radosevic et al., OCEANS 2011, JOE 2014; Tian et al., OCEANS 2026; Wang, OCEANS 2022; Jing, IoT-J 2025 (delay-Doppler)	Predict the channel, not the motion; MCS or bit loading
Doppler-indexed AMC	Park et al., Remote Sens. 2026 (MDPI, not in library) SOURCE	Doppler as a table index; MCS only
Feedback timing	AoI-aware hierarchical bandit (Busacca et al., GLOBECOM 2025)	No physical model
Error control under delay	OLLA (LTE/5G); ACI (Gibbs–Candès 2021); delayed ACI (El Halabi–Brandt 2026)	Same update rule as our margin; new: action-dependent delay, service accounting
AMC survey and benchmark	Cooper-Baldock et al., Appl. Ocean Res. 2026 (open-data benchmark)	Baselines to compare against

Differences read from abstracts; full scope still to be checked. SOURCE: found on the web only.

Related work (2): Doppler is compensated, and packets are designed offline

Line of work	Representative papers	Use of Doppler
Physics of wave-induced Doppler	Walstead, JASA 2016–17; Rawat, IEEE TAES 2021 (Doppler spectrum from sea-surface motion); Li et al., JOE 2008 (residual offset varies with waves)	Characterisation; design of compensation
Packet structure for a motion range	Zakharov et al., JOE 2019 ($\lvert v\rvert \leq 5\,\mathrm{m/s}$, $\lvert a\rvert \leq 1\,\mathrm{m/s^2}$, $\approx$1 s packets)	Fixed offline design, not adapted online
Acceleration-aware compensation	Abdelkareem et al., Ad Hoc Netw. 2015; Cho and Kim, IET Commun. 2026; Kalman Doppler tracking (Hu et al., MSN 2024, FH-FSK)	Receiver absorbs constant drift; residual comes from jerk
Doppler prediction in the receiver	Ceballos Carrascosa and Stojanovic, JOE 2010 (Doppler predicted across OFDM blocks)	In the receiver; not used by the transmitter
Two-way Doppler	UWGS time sync (2024); geo-Doppler ranging (OCEANS 2026)	Timing, position

Enablers now public: Underwater Acoustic Channel Library (Li et al., 2026; bulk drift kept, one-way, $\approx$50–60 s records), Qarabaqi–Stojanovic simulator, NOAA NDBC wave spectra. Caution: Watermark may remove bulk Doppler.

Candidate gaps, re-ranked after the library check

G1 Pilot layout and packet length are fixed offline for an assumed motion range

Zakharov et al. 2019, Hong et al. 2025, acceleration-aware receivers: designed once for $|a| \leq 1\,\mathrm{m/s^2}$.

Pilot design $\times$ Doppler: 4 papers, none forecast-driven (closest: Van der Werf et al. 2024, OTFS).

G2 Receivers re-estimate Doppler on every packet, without a kinematic prior

Closest: Doppler predicted across OFDM blocks inside the receiver (Ceballos Carrascosa–Stojanovic 2010).

G3 Doppler measured at both ends is not fused for the link

Two-way Doppler serves time sync and ranging (UWGS 2024, geo-Doppler 2026) only.

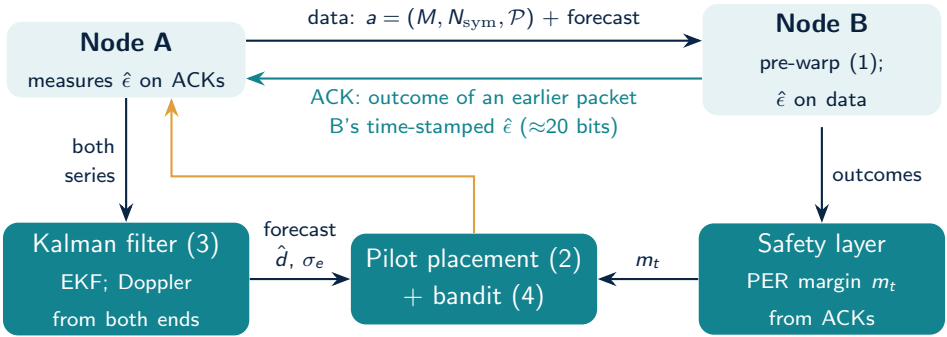
G4 Learning link adaptation ignores kinematic context and arm-dependent delay

UWA contextual bandits: 3 papers, band or relay selection (Schutz et al. 2023); AMC learners use no kinematic context.

Dropped: send timing to the swell (high risk for acceleration-aware receivers). Counts: regex on title+abstract, cat. 01–13, 8 Oct 2026.

Proposed methodology

System at a glance



Numbers in brackets: the four blocks on the following slides. OPEN For the technical specification: header fields and quantisation; out-of-sequence measurements; SNR source; NACK or timeout, packet matching, ACK vs data loss; whether $\hat{\epsilon}$ needs payload decoding or only preamble acquisition.

Two quantities that must not be confused

Doppler scale (dimensionless)

$$\epsilon(t) = \frac{v_{\mathrm{rel}}(t)}{c}$$

Physical Doppler drift (units s^{-1})

$$d(t) = \dot{\epsilon}(t) = \frac{\dot{v}_{\mathrm{rel}}(t)}{c}$$

Some papers call ϵ the "Doppler rate". Here d is always the *physical* drift, never the receiver's residual.

- What harms a packet is what remains after compensation: it depends on receiver and action, not on d alone: forecast error $e \to$ receiver, action $a \to$ residual impairment $\phi_a(e) \to$ PER
- Example: $|d| = 10^{-4}\,\mathrm{s}^{-1}$ is a relative acceleration of $0.15\,\mathrm{m/s}^2$ at $c = 1500\,\mathrm{m/s}$.
- One common (bulk) component is an initial restriction. If path powers, delay spread, differential Doppler or tracking state change PER at equal d, that is a model limitation OPEN.

Model

Motion CLAIM: $v_{\mathrm{rel}}(t)$ = slow platform drift + damped oscillators (swell, wind sea) learned online.

Physical drift $d = \dot{v}_{\mathrm{rel}}/c$ in s^{-1}, common (bulk) component only: an initial restriction OPEN.

Observations: time-stamped $\hat{\epsilon}$ on the preamble of every packet received at *either* end (accuracy set by preamble length and bandwidth); ACK or timeout for earlier packets.

Forecast error $e(u) = d(u) - \hat{d}(u)$ over the receive interval, standard deviation $\sigma_e(u)$ from the estimator. A receiver without the prior has $\hat{d} = 0$, so $e = d$ (the baseline).

Failure CLAIM: the receiver turns e into a residual impairment $\phi_a(e) \geq 0$, depending on $(M, N_{\mathrm{sym}}, \mathcal{P})$; a monotone scalar curve is a hypothesis, checked by calibrating each receiver in simulation:

$$\mathrm{SNR}_{\mathrm{eff}} = \frac{\mathrm{SNR}}{1 + \mathrm{SNR}\,\phi_a(e)} < \gamma_a \iff \phi_a(e) > \frac{1}{\gamma_a} - \frac{1}{\mathrm{SNR}}, \qquad e_{\mathrm{tol}}(a, \mathrm{SNR}) = \sup\left\{|e| : \phi_a(|e|) \leq \tfrac{1}{\gamma_a} - \tfrac{1}{\mathrm{SNR}}\right\}$$

Empty set (e.g. $\mathrm{SNR} < \gamma_a$): a excluded; unbounded: $e_{\mathrm{tol}} = \infty$; non-monotone ϕ_a: use the set itself.
γ_a: SNR at which a meets the PER target, a measured threshold SUGGESTED.

Traffic: bursty arrivals; waiting $h \leq h_{\max}$ set by the MAC; a delay budget per packet.

A minimal action space SUGGESTED

$$a = (M,\ N_{\mathrm{sym}},\ \mathcal{P})$$

- M: modulation order, QPSK or 16-QAM
- N_{sym}: OFDM symbols in the packet
- $\mathcal{P} \subset \{1, \ldots, N_{\mathrm{sym}}\}$: positions of full training symbols, $N_{\mathrm{tr}} = |\mathcal{P}|$

The receiver pre-warp is not an action: it always follows the forecast. Fixed in the first paper: useful symbol duration T_s, cyclic prefix T_{cp}, code. Waiting time h: set by the MAC.

Packet duration

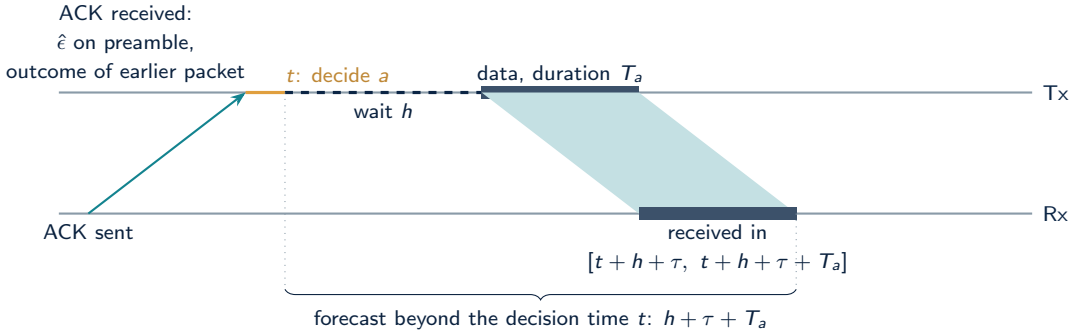
$$T_a = T_{\mathrm{pre}} + N_{\mathrm{sym}}\,(T_s + T_{cp})$$

Payload

$$B_a = \left(N_{\mathrm{sym}} - |\mathcal{P}|\right) N_{\mathrm{sc}}\, \log_2 M\, R_c$$

Uniform $\mathcal{P}$ with spacing K_{upd} is the baseline. Exact count from the frame (first estimate from the preamble? final training symbol?) OPEN. B_a: information bits before header, CRC, termination; split messages count every segment and its delay.

When the packet is received, and what is known then



τ: one-way propagation time ($\approx$0.67 s/km). t_{obs}: epoch of the physical state that $\hat{\epsilon}$ represents, fixed by the measurement equation OPEN. From an estimate at t_{obs} the horizon is $H = (t - t_{\mathrm{obs}}) + h + \tau + T_a$; if the filter is already propagated to t, $h + \tau + T_a$ remains. The ACK of this packet arrives $\geq h + 2\tau + T_a$ after the decision, so the margin reacts to old outcomes.

Shorter packets or more frequent tracking?

Doppler perfectly corrected at the start of an interval, then held fixed; roughly constant drift, narrowband around f_c:

$$|\Delta f(T)| \approx f_c\,|d|\,T \quad \Rightarrow \quad T \leq \frac{\Delta f_{\max}}{f_c\,|d|}$$

T is the interval without a compensation update:

- blockwise receiver: $T =$ whole packet $\Rightarrow$ shorten N_{sym}
- tracking receiver: $T = T_{\mathrm{upd}} = K_{\mathrm{upd}}(T_s + T_{cp})$ $\Rightarrow$ reduce K_{upd}, keep N_{sym}
- forecast-aided (Block 1): $|d| \to |d - \hat{d}|$

Illustration only SUGGESTED

$f_c = 20\,\mathrm{kHz}$, assumed $\Delta f_{\max} = 2\,\mathrm{Hz}$:
$|d| = 10^{-4}\,\mathrm{s}^{-1} \Rightarrow T \leq 1\,\mathrm{s}$
$|d| = 2 \times 10^{-4}\,\mathrm{s}^{-1} \Rightarrow T \leq 0.5\,\mathrm{s}$

$\Delta f_{\max}$ is not a measured receiver threshold.

Left out: wideband resampling effects, differential path Doppler, accumulated phase error, estimator noise, the actual detector. Valid only for fixed correction and roughly constant drift: it explains the mechanism, it is not a reliability guarantee.

Block 1: receiver pre-warped along the forecast SUGGESTED

Motion time-warps the received signal: $r(t) \approx s\big(t + \int_0^t \epsilon(u)\,du\big)$.

Today: estimate ϵ on the preamble (constant, or constant + acceleration), resample, track.

Here: resample along the forecast $\hat{\epsilon}(t)$ over the whole packet; the tracker sees only the residual warp $\int(\epsilon - \hat{\epsilon})$:

$$|\Delta f(T)| \approx f_c\,|d - \hat{d}|\,T \quad \Rightarrow \quad T \le \frac{\Delta f_{\max}}{f_c\,|d - \hat{d}|}$$

Same simplified model as before, with the drift replaced by the forecast error. Closest: Ceballos Carrascosa–Stojanovic, JOE 2010.

What it buys

- fewer pilots at equal PER, or longer packets, even if acceleration-aware receivers absorb the drift
- PER now depends on the forecast uncertainty σ_e, which the estimator reports

What it needs OPEN

- the forecast at node B: a few header bits, or the same filter running there
- a fallback if the forecast is wrong: residual test, preamble-only

“More pilots” made explicit

Five different things are called pilots:

1. more frequent temporal tracking observations
2. more pilot subcarriers inside one OFDM symbol
3. periodic full training symbols or midambles
4. known preambles at packet boundaries
5. decision-directed observations, if the receiver supports them

Against *temporal* drift the variable that matters is the spacing between usable tracking observations, K_{upd}.

Baseline: uniform spacing $K_{\mathrm{upd}} \in \{8, 4, 2\}$, test points, not thresholds. Proposed: a uniform grid at the largest spacing K_{max} that channel estimation allows, plus extra positions from the forecast uncertainty (Block 2) SUGGESTED. Per-symbol tracking needs embedded pilots and its own overhead model OPEN.

An observation and a receiver update are not the same operation: each receiver must state its estimator and what it needs from an observation OPEN.

Block 2: pilots where the forecast is uncertain SUGGESTED

The estimator predicts how the error variance $\sigma_e^2(u)$ grows over the packet, $u \in [0, T_a]$; each training symbol resets it locally.

Placement problem

$$\min_{\mathcal{P} \supseteq \mathcal{P}_0:\ |\mathcal{P}| \le N_{\mathrm{tr}}} \int_0^{T_a} \sigma^2_{e|\mathcal{P}}(u)\, du$$

$\mathcal{P}_0$: uniform grid at the largest spacing $K_{\max}$ the tap coherence time allows, so equalisation never loses the channel; the forecast adds only the extra pilots. Baseline: $\mathcal{P} = \mathcal{P}_0$.

OPEN $K_{\max}$ is receiver-dependent (comb pilots relax it; $K_{\max} \ge N_{\mathrm{sym}}$: $\mathcal{P}_0 = \emptyset$, the unconstrained problem); integral or max over u; submodularity for greedy (Krause et al. 2008).

Expected structure CLAIM (to derive)
For a swell $d = A\sin(\omega t + \varphi)$, a phase error $\delta\varphi$ gives a drift error $A\,\delta\varphi\cos(\omega t + \varphi)$: largest where d crosses zero, where the jerk peaks. Phase error grows with the horizon, so it dominates.

$\Rightarrow$ extra pilots cluster around the zero crossings of d. The jerk, which hurts acceleration-aware receivers most, decides where the extra pilots go.

Theory target (R3): closed-form placement for one swell component; greedy with a bound in general; overhead saved vs uniform spacing.

Block 3: both ends measure Doppler SUGGESTED

Node B measures $\hat{\epsilon}$ on each data packet, node A on each ACK. Both see the same range rate, so a Kalman filter on the drift-plus-oscillators model fuses the two interleaved series (an EKF while the sea parameters are learned) SUGGESTED.

Largest gap between samples (stop-and-wait):

$$\Delta_{\text{one-way}} \geq 2\tau + T_a, \qquad \Delta_{\text{two-way}} \approx \tau + T_a$$

No aliasing for a swell of period P ($\Delta < P/2$):

$$R < \tfrac{c}{2}\left(\tfrac{P}{2} - T_a\right) \;\longrightarrow\; R < c\left(\tfrac{P}{2} - T_a\right)$$

$P = 8\,\text{s}$, $T_a = 1\,\text{s}$: $2.25\,\text{km} \rightarrow 4.5\,\text{km}$.

Cost and open points OPEN

- B's estimate returns in the ACK header: value + time stamp, ≈20 bits SUGGESTED
- it arrives late: out-of-sequence measurements in the filter
- holds both ends' motion projected on the path, unlike one IMU
- assumes both directions share the bulk Doppler (to check in calibration)

Theory target (R2): observability and forecast error vs range, one- vs two-way sampling.

Block 4: a contextual bandit that knows the physics SUGGESTED

Context x_t: forecast over the candidate receive interval ($\hat{d}$, σ_e), $\widehat{\mathrm{SNR}}$, range.

Arms $a = (M, N_{\mathrm{sym}})$; for each arm, $\mathcal{P}$ comes from Block 2.

Reward: delivered bits per unit airtime, $B_a \, \mathbf{1}\{\text{success}\} / T_a$ (one objective, fixed before the comparison).

Model: failure probability monotone in forecast error and action; generalised linear in physical features,

$$p_{\mathrm{fail}}(a, x) = g\big(\theta_a^{\top} \psi(x)\big),$$

θ_a initialised from the calibration curves, learned online (UCB or Thompson sampling).

What is new for the theory (R4)

- feedback for arm a arrives $D_a \geq h + 2\tau + T_a$ later: the delay depends on the arm
- short packets return feedback sooner, biasing a naive learner towards them OPEN
- target CLAIM: regret $\tilde{O}(\sqrt{T})$ + delay term, extending fixed-delay results (Joulani et al., ICML 2013)

Safety layer: only arms meeting the PER target with margin m_t.

Builds on AMUSE, POSEIDON, AoI bandit (Busacca et al.).

Safe set and decision, written out

A threshold rule as a constraint. The learner plays only arms in the safe set

$$\mathcal{S}_t = \left\{ a : \ \hat{p}^{\,\mathrm{up}}_{\mathrm{fail}}(a, x_t) \le \alpha - m_t \right\}$$

$\hat{p}^{\,\mathrm{up}}$: upper confidence bound of the bandit model; α: PER target; m_t: margin from delayed ACKs (theoretical objectives). Without the prior and with a deterministic threshold this reduces to a plain threshold rule, predicted drift + margin $\le d_{\mathrm{tol}}$.

The bound must hold over the *whole* receive interval: a pointwise quantile does not bound the maximum. Tracking may make cumulative phase error the better statistic; calibration decides.

OPEN $\widehat{\mathrm{SNR}}$: who measures it, which link, feedback bits, age at reception. In first simulations it is an oracle.

Procedure per packet SUGGESTED

1. fuse both ends' time-stamped Doppler
2. forecast $\hat{d}$, σ_e per receive interval
3. place pilots for each arm (Block 2)
4. safe set from predicted PER and m_t; if empty: fallback, wait or drop
5. the bandit picks an arm in it
6. send frame + forecast; B pre-warps
7. update θ, m_t from delayed ACKs

Theoretical objectives

Theory: four core results, the rest as extensions

R1 Predictability

How long Doppler stays predictable (L1; idea document: $\approx$13 s for an 8 s swell vs $\approx$1 s for fading); mixed seas degrade gracefully (P2)

R2 Sampling

Forecast error vs range, one- vs two-way ($\times 2$ samples per round trip); observability with delayed, irregular samples (L3–L4)

R3 Pilots

Optimal extra positions on top of the grid $\mathcal{P}_0$ for one swell component; greedy with a bound in general (from P7)

R4 Learning

Regret of the physics-informed bandit with arm-dependent feedback delay (from T12)

Supporting: T10–T11, PER safety layer and its stability (known structure: OLLA / ACI).
Extensions: T8 quiet windows, T9 range of Doppler vs SNR feedback, L3–L4 path separation, P5 safe identification, P6 optimal action.

OPEN For each result: proved, partial proof, conjecture or planned? Not yet listed. Known hard parts: out-of-sequence samples in R2, submodularity under the grid in R3, arm-dependent delay in R4.
L/P/T: lemma, proposition, theorem of the idea document.

Safety layer: what makes the PER claim non-vacuous

Deck claim CLAIM: long-run PER below target for any channel, predictor and estimate.

Margin update. The deck's verbal rule, written as

$$m_{k+1} = m_k + \eta\,(e_k - \alpha), \quad e_k = \mathbf{1}\{\text{failure}\}$$

per processed outcome (α: PER target). Unprojected, $\frac{1}{N}\sum_k e_k = \alpha + \frac{m_{N+1} - m_1}{\eta N}$: calibration only if m grows sublinearly (if all fail, it does not). OPEN Signed correction or buffer $m \geq 0$? (projection changes the proof); outcome vs packet index.

Which guarantee? Long-run frequency, finite-horizon, or per-packet. Report it with goodput, drops, deadline misses, retransmissions.

To fix before proving Thm. 10 OPEN

- PER denominator: packets sent? with retransmissions? deadline expiries?
- missing ACK or timeout: failure, or excluded?
- empty feasible set: fallback action, wait, or drop, and how each is counted
- minimum service required, e.g. a bound on the dropped or delayed fraction
- feasibility assumption, e.g. some action always meets the target with slack
- action-dependent delay: outcomes arrive $\geq h + 2\tau + T_a$ late

Questions for the group

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- Is the scope right for one paper, or should the bandit (Block 4) be a follow-up?
- Can the target modem or simulator carry $\approx$20 header bits for the forecast and the far end's Doppler?
- What is already implemented, and which results are proved?
- May the policy abstain, drop packets or use a fallback, and how is service counted?
- How much time and theory/simulation support is available?
- First test: does a forecast prior still help acceleration-aware receivers, in PER or in pilots saved?